The Infinite Brocade: A Corpus and Critical Exegesis of Su Hui's *Xuanji Tu*

I. Introduction: The Universe in a Square of Silk

The *Xuanji Tu* (璇璣圖), widely known in English as the *Picture of the Turning Sphere* or the *Star Gauge*, stands as a singular anomaly in the history of world literature. Composed in the fourth century CE by the poet Su Hui (courtesy name Ruolan), it is ostensibly a poem of complaint and longing addressed to an estranged husband. However, to categorize it merely as a "poem" is a category error. It is, more accurately, a computational engine made of silk and language—a 29 by 29 character grid containing 841 logograms that can be read in thousands of directions to generate a potentially infinite series of rhyming verses.

This document serves as a comprehensive corpus and analytical report on the *Xuanji Tu*. It aggregates historical data, textual reconstruction, algorithmic analysis, and literary criticism to provide an exhaustive account of the work. The *Xuanji Tu* represents the apex of the *huiwen* (circular or reversible) genre, surpassing all predecessors and successors in complexity. While the original textile artifact—an eight-inch square of polychrome brocade—has been lost to the ravages of the intervening sixteen centuries, the text survives through a lineage of manuscript transmission, scholarly reconstruction, and legendary accretion.

The significance of the *Xuanji Tu* extends beyond its status as a literary curiosity. It serves as a profound philosophical treatise on the nature of the cosmos, modeled on the armillary sphere (*xuanji*) used by ancient Chinese astronomers to map the heavens. Just as the stars revolve around the pole, the characters of the poem revolve around a central void (later filled by the character *Xin* or Heart), creating a text that has no absolute beginning or end, only an eternal, cyclical recurrence of sorrow and redemption. This report dissects the architecture of this "paper machine," exploring how Su Hui utilized the binary constraints of the loom and the grid to achieve a freedom of expression that borders on the infinite.

II. Historical Context: The Weaver and the Era of Fragmentation

2.1 The Sixteen Kingdoms and the Collapse of Order

To understand the genesis of the *Xuanji Tu*, one must situate it within the chaotic milieu of the Sixteen Kingdoms period (304–439 CE). Following the collapse of the Western Jin dynasty, northern China fractured into a kaleidoscope of short-lived non-Han regimes. Su Hui lived under the Former Qin (351–394 CE), a state established by the Di ethnic group, which briefly unified northern China under the rule of Fu Jian.

This era was characterized by:

- **Political Instability:** Regimes rose and fell with violent rapidity.
- **Cultural Hybridity:** The influx of non-Han peoples and the rapid spread of Buddhism (particularly the Mādhyamaka school) created a fertile ground for intellectual experimentation.

- **The Preeminence of the Literati:** Despite the chaos, the mastery of classical forms remained the primary currency of the elite.

Su Hui was born into this world of flux. She hailed from a prestigious family in Wugong (modern Shaanxi); her father, Su Daozhi, served as the magistrate of Chen Liu. This aristocratic background granted her access to literacy, a privilege denied to the vast majority of women in early medieval China. Historical sources describe her as intellectually precocious, possessing a "brilliant and effective" mind capable of composing thousands of literary works, though nearly all were lost save for the *Xuanji Tu*.

2.2 The Narrative of Estrangement: History vs. Hagiography

The textual history of the *Xuanji Tu* is inseparable from the biographical legend of its creation. The narrative acts as the "operating instructions" for the machine, explaining *why* such a complex object was brought into existence.

The Core Conflict: Su Hui was married to Dou Tao, a general and administrator for the Former Qin. The central conflict arises from Dou Tao's relationship with a concubine, Zhao Yangtai. When Dou Tao was appointed governor of Qinzhou (in present-day Gansu) and subsequently exiled to the desert frontier (often identified as Dunhuang or the Tartar region), he intended to take his concubine with him.

The Divergence of Accounts:

- **The Jealous Wife:** Early sources frame Su Hui's refusal to accompany Dou Tao as an act of jealousy. She refused to share his affections with Zhao Yangtai, staying behind in the central plains while he traveled to the frontier.
- **The Betrayal:** Dou Tao, in violation of a vow of fidelity, took the concubine with him, effectively abandoning Su Hui.
- **The Repentance:** Left in isolation, consumed by *si* (longing/thought), Su Hui channeled her grief into the production of the *Xuanji Tu*. She wove the poem into a brocade and sent it to him.

The reception of the object by Dou Tao is the climax of the legend. It is said that the complexity of the poem—requiring immense intellect to decode—served as proof of her superior mind and depth of feeling. Upon unraveling the text (and presumably realizing that the concubine could offer no such intellectual companionship), Dou Tao sent Zhao Yangtai away and welcomed Su Hui back with elaborate ceremony.

Sociological Implications: This narrative subverts the traditional "boudoir lament" (*guiyuan*). Usually, the abandoned woman is passive, fading away in silence. Su Hui, however, takes active control. She creates a *weapon* of text. The complexity of the grid forces the reader (the husband) to engage with her mind for hours, perhaps years. As Su Hui herself reportedly wrote: "one lingers endlessly in the poem, twisting and turning". The poem entraps the husband in a labyrinth of her own making, enforcing a mental communion that physical distance had severed.

III. The Architecture of the *Xuanji Tu*: Object and Techne

3.1 Materiality: The Lost Brocade

Before it was a text, the *Xuanji Tu* was a textile. The transition from *text* (writing) to *textile*

(weaving) is crucial. The etymology of "text" (from Latin *texere*, to weave) is literalized here.
Specifications of the Artifact:

Attribute	Description	Source
Dimensions	8 inches by 8 inches (approx. 20 cm x 20 cm)	
Material	Silk Brocade (*Jin*)	
Technique	Shuttle-woven (possibly embroidered in later reproductions)	
Colors	Five (Red, Yellow, Blue, White, Purple/Black)	
Grid Size	29 x 29 characters (841 total)	

The Significance of the Loom: Weaving is a digital process. The warp and weft intersect at coordinate points (x, y). By composing directly on the loom, Su Hui accepted the strict grid as a constraint. Unlike a brush which can flow and cursively link characters, the loom isolates each character in its own cell. This isolation is what permits the multidirectional reading; there is no "ligature" of handwriting to enforce a left-to-right flow. The characters function as movable type centuries before Gutenberg, held in place by silk threads.

3.2 The *Xuanji* Metaphor: The Armillary Sphere

The title *Picture of the Turning Sphere* is not poetic flourish; it is the structural logic of the piece. The *Xuanji* (armillary sphere) was the primary astronomical instrument of the time, consisting of nesting rings (meridians) representing the celestial equator, the ecliptic, and the horizon.
Cosmological Mapping:

- **The Center:** The pole star (North Star) or the Earth, around which the cosmos turns. In the poem, this is the character *Xin* (Heart/Mind).
- **The Rings:** The color-coded regions of the poem correspond to the rings of the sphere. The "Red Armature" likely represents the mobile rings of the instrument, allowing the reader to "turn" the text.
- **The Infinite Cycle:** Just as the heavens have no beginning or end, the circular poems on the perimeter have no start or stop. The reader can enter the stream at any point.

This mapping elevates the domestic tragedy of a marriage to a cosmic event. The wife's heart becomes the axis mundi; the husband's departure is a planetary misalignment that must be corrected through the "gravity" of the poem.

3.3 The Five Colors and *Wuxing*

The five colors are not merely decorative; they serve as the "operating system" or the "legend" for the map. They correspond to the *Wuxing* (Five Elements) theory, which governed Chinese medicine, astronomy, and politics.

Color	Element	Direction	Region in Poem	Reading Rule Characteristics
Red	Fire	South	The Armature (Grid lines)	7-character segments; turns at junctions
Yellow	Earth	Center	Central blocks	5-character lines;

Color	Element	Direction	Region in Poem	Reading Rule Characteristics
				spiral readings
Blue/Green	Wood	East	Corners	3-character lines forming 6/12 line poems
White	Metal	West	(Specific blocks)	4-character or 6-character variants
Black/Purple	Water	North	(Specific blocks)	4-character or 6-character variants

Note: Sources vary slightly on the exact color-to-region mapping (e.g., Blue vs. Green, Black vs. Purple), but the principle of Five Colors regulating the reading path is consistent.

IV. The Textual Corpus: Data and Exegesis

Using the reconstructed text provided in the research material , we can analyze the specific linguistic data of the grid. The poem is a dense matrix of classical Chinese, characterized by a lack of inflection and a high density of content words.

4.1 The Outer Circle (The Perimeter)

The outermost border (Rows 1 and 29, Columns 1 and 29) forms the famous circular poem. This is often the entry point for the reader.

Row 1 (Top Edge):

琴清流楚激弦商秦曲发声悲摧藏音和咏思惟空堂心忧增慕怀惨伤仁 *Qin qing liu Chu ji xian Shang Qin qu fa sheng bei cui cang yin he yong si wei kong tang xin you zeng mu huai can shang ren*

Analysis: This line establishes the musical theme.

- *Qin* (Zither) and *Shang* (Musical note/mode associated with autumn and sorrow).
- *Chu* and *Qin* (State names, evoking ancient songs).
- *Keywords:* "Sad" (*bei*), "Heartbreak" (*cui*), "Empty Hall" (*kong tang*).
- *Translation:* "The zither's clear flow, the Chu songs arouse the Shang strings; Qin tunes emit sounds, sad and heartbreaking, hidden tones harmonize the chant; thinking only of the empty hall, the heart's worries increase, longing and cherishing, miserable and wounded benevolence."

Row 29 (Bottom Edge):

親剛柔有女為賤人房憂處已憫薇身長路悲曠感士民梁山殊塞隔河津 *Qin gang rou you nü wei jian ren fang you chu yi min wei shen chang lu bei kuang gan shi min Liang shan shu sai ge he jin*

Analysis: This line shifts to the social reality of the abandoned woman.

- *Jian ren* (Lowly person): The wife refers to herself humbly, or perhaps refers to the concubine (depending on reading direction).
- *Liang Shan* (Mount Liang): A geographical barrier separating the lovers.
- *Ge he jin* (Separated by river and ford): The motif of the obstacle.

4.2 The Central Axis (The Spine)

Row 15 and Column 15 intersect at the center.
Row 15 (Horizontal Axis):
新舊間離天罪辜神恨昭感興作蘇心璣明別改知識深微至嬖女因佞臣 *Xin jiu jian li Tian zui gu Shen hen zhao gan xing zuo Su Xin Ji ming bie gai zhi shi shen wei zhi bi nü yin ning chen*
Critical Exegesis: This line is the "credits sequence" and the "source code" of the poem.

- **Su Xin Ji (蘇心璣):** The characters Su (Surname), Xin (Heart), and Ji (Sphere/Mechanism) appear sequentially. This is a signature: "Su's Heart Sphere" or "The Sphere of Su's Heart."
- **Narrative of Creation:** "Arise make" (*xing zuo*) suggests the creation of the poem.
- **The Cause:** "Favorite concubine" (*bi nü*) and "Flatter minister" (*ning chen*). The line explicitly blames the "favorite woman" and the "flattering officials" (perhaps those who led her husband astray) for the separation.

4.3 The Four Corners (The Blue Regions)

The corners contain dense blocks of characters that form shorter poems, often with different prosodic rules (e.g., 3-character lines).
Top Right Corner (3x3 Block Extraction): Characters here are arranged to allow reading in zigzag patterns.

- *Insight:* "The top right corner in blue... is made up of 3-character lines... poem length is either 6 or 12 lines."
- *Reading Path:* Read horizontally (chars 1-3), drop down, read backward (chars 6-4).
- *Rhyme:* The rhyme scheme must hold the vertical and horizontal readings together.

4.4 The Full Grid Corpus (Row-by-Row Data)

To provide the "Corpus" requested, below is a structured representation of key rows from the grid.

Row	Text Extract (First 10 Characters)	Thematic Focus
01	琴清流楚激弦商秦曲发	Music, Sorrow, Sound
02	房廂東步階西遊王姿淑	Domestic Space (West/East Room), Beauty
03	蘭休桃林陰翳桑懷歸思	Nature (Peach/Mulberry), Longing to Return
04	凋翔飛燕巢雙鳩士迤逶	Birds (Swallows/Doves), Flight, Departure
05	茂流泉清水激揚思欣其	Water (Springs/Streams), Clarity, Emotion
...	...	...
15	新舊間離天罪辜神恨昭	**Axis:** Separation, Sin, Su Hui's Name
...	...	...
29	親剛柔有女為賤人房憂	Social Status, Distance,

Row	Text Extract (First 10 Characters)	Thematic Focus
		Geography

Insight: The grid moves from the "high" imagery of music and nature (Top) to the "heavy" imagery of geography and social status (Bottom). The center (Row 15) acts as the pivot between the celestial/abstract and the earthly/concrete.

V. Mechanics of Reading: The Algorithm of the Grid

The *Xuanji Tu* anticipates the logic of hypertext and database aesthetics. It is a text that is not meant to be read linearly but *navigated*.

5.1 The Logic of *Huiwen* (Reversibility)

The Classical Chinese language is uniquely suited for this architecture due to its isolationist morphology.

- **No Inflection:** A character does not change form for tense or number.
- **Fluid Syntax:** SVO (Subject-Verb-Object) is standard, but OVS or VSO can be poetic variants.
- **Polysemy:** A character like *Shang* (傷) can mean "wound" (noun), "to hurt" (verb), or "sad" (adjective).

Example of Reversibility :

- *Forward:* "The house no longer has a master." (*Tang kong wei si*)
- *Reverse:* "You no longer have a home." (or "The empty hall thinks only of...") The reversal shifts the agency. In the forward reading, the house is empty because he left. In the backward reading, *he* is homeless because he abandoned the domestic center.

5.2 The Red Armature: A Decision Tree

The "Red" region functions as a grid of 7-character segments.

- **The Junctions:** At the intersection of the red lines, the reader faces a choice.
- **Combinatorics:** If there are 32 segments, and each segment connects to 3 others, the potential paths expand factorially.
- **User Agency:** This makes the reader a co-author. The emotion of the poem depends on the path chosen. A path turning "left" might emphasize anger; a path turning "right" might emphasize sorrow.

5.3 Algorithmic Decoding (Modern Analysis)

In the Ming Dynasty, scholars found 7,940 readings. Modern computational analysis suggests the number is much higher, potentially infinite if recursive looping is allowed.

- **The Star Gauge Project / Jen Bervin:** Modern projects treat the poem as a dataset. The generated_segments.csv mentioned in snippet reveals that modern code is required to map what Su Hui held in her mind.
- **Graph Theory:** The poem is a graph G = (V, E), where V are the characters and E are the valid syntactic connections allowed by the color rules. A poem is a walk on this graph.

5.4 Oulipo and the Prehistory of Potential Literature

The *Xuanji Tu* is often compared to the work of the Oulipo group (e.g., Raymond Queneau's *Cent mille milliards de poèmes*).

- **Comparison:** Queneau used 10 sonnets with interchangeable lines (10^{14} combinations). Su Hui used a single grid where the *direction* changes the line itself.
- **Constraint:** Both use strict constraints (sonnet form vs. rhyme/grid) to generate freedom. Su Hui's work is arguably more complex because it lacks the "strips" of Queneau; every character is structurally load-bearing for multiple crossing poems.

VI. Hermeneutics: Themes and Philosophy

6.1 The Poetics of Obsession

The form of the poem mirrors the psychological state of the author. *Si* (longing) is obsessive; it circles the same thoughts repeatedly.

- **The Labyrinth:** By trapping the husband in the grid, Su Hui forces him to experience the claustrophobia of the boudoir. He cannot just "read" the letter and put it away; he must struggle with it, just as she struggled with his absence.
- **The Infinite Loop:** The circularity implies that the grief has no end. The only way to break the loop is for the husband to physically return.

6.2 Time and Entropy

The *Xuanji* (Armillary Sphere) measures the movement of the heavens, which is the movement of time.

- **Reversing Time:** If the poem can be read backwards, it suggests a desire to rewind time—to go back to the moment before the betrayal. The poem is a "time machine" attempting to reverse the entropy of the relationship.
- **Seasons:** The poem is saturated with seasonal imagery (Spring flowers, Winter pines). These mark the passage of time during his absence.

6.3 The Political Allegory: Loyalty (*Zhong*)

In Chinese poetic tradition, the relationship between husband and wife often mirrors that between ruler and minister.

- **The Loyal Minister:** Su Hui compares herself to the "loyal minister" (*chen*) in the central row. By remaining faithful despite "exile" (abandonment), she claims the moral high ground.
- **The Tyrant/Bad Ruler:** Dou Tao, by listening to the "flattery" of the concubine, acts as a deluded ruler. The poem is a "remonstrance" (*jian*), a formal duty of the loyal subordinate to correct the superior.

VII. Transmission and Reception History

7.1 The Tang Dynasty: Empress Wu's Endorsement

The Tang dynasty saw a revival of the *Xuanji Tu*, spearheaded by Empress Wu Zetian.

- **Wu Zetian's Preface:** The Empress wrote a preface to the poem, celebrating Su Hui's genius. Wu Zetian likely saw in Su Hui a reflection of her own struggle for female legitimacy in a male-dominated sphere. The story of Su Hui became a "founding myth" for female literary talent.

7.2 The Song Dynasty Gap

During the Song, the text became scarce. The physical brocade was lost. The "Heart" character controversy began here—was the center originally empty?

- **The Void:** Daoist philosophy emphasizes the utility of the void (the empty hub of the wheel). An empty center in the *Xuanji Tu* is philosophically potent. It represents the *Dao*—unnamed and formless—around which the "ten thousand things" (the characters) revolve. The later addition of *Xin* (Heart) makes the metaphor literal but loses the Daoist nuance.

7.3 Ming and Qing Decoders

The Ming dynasty saw a surge in "puzzle solving." Scholars like Wan Shu published treatises mapping the specific reading paths. This era treated the poem less as an emotional cry and more as a marvelous object or curio.

7.4 Contemporary Digital Art

Today, the *Xuanji Tu* is finding new life in the digital humanities.

- **Jen Bervin:** Her work *Su Hui's Picture of the Turning Sphere* (2016-2020) rematerializes the poem. She emphasizes the "feminist listening room"—a space where the sheer volume of Su Hui's voice (12,000 poems) can finally be heard.
- **The App:** The shift from silk to touchscreen is seamless. Both interfaces are tactile and grid-based. The *Xuanji Tu* was, effectively, the first "app" designed to run on the hardware of the human mind.

VIII. Conclusion: The Unending Poem

The *Xuanji Tu* stands as a monument to the capacity of the human mind to find infinite freedom within the strictest of constraints. Su Hui, confined by the walls of her gender and the geography of her abandonment, constructed a universe that transcended both.

This corpus analysis reveals that the *Xuanji Tu* is not merely a "poem" in the Western sense of a linear lyrical expression. It is a **generative system**. It utilizes the technology of its time (the loom, the armillary sphere, the five elements) to build a machine capable of processing grief into beauty.

The return of Dou Tao, often dismissed as a romantic legend, can be understood as the inevitable result of the text's power. Faced with an object of such overwhelming complexity and devotion—a text that required the rest of his life to fully read—he had no choice but to recognize the superior gravity of the wife he had left behind. The *Xuanji Tu* did not just communicate a

message; it restructured the reality of its recipient. It remains, sixteen centuries later, a "star gauge" that measures the depth of human longing against the cold turning of the cosmos.

IX. Appendix: Technical Data Tables

Table 1: Structural Parameters of the *Xuanji Tu*

Parameter	Value	Notes
Grid Dimensions	29 x 29	Perfect Square
Total Cells	841	29^2
Central Cell	(15, 15)	Originally Empty or containing "Xin"
Character Set	~800 unique	High lexical density; few particles
Colors	Red, Yellow, Blue, White, Purple	Mapping to Wuxing (Five Elements)
Possible Readings	3,000 - 12,000+	Depending on strictness of constraints

Table 2: The Five Regions and Their Rules (Reconstructed)

Region Name	Color	Location	Line Length	Reading Direction
The Armature	Red	Grid Overlay	7 chars	Linear, turns at junctions
The Earth Square	Yellow	Middle Ring	5 chars	Spiral, block reading
The Corners	Blue	4 Corners	3 chars	Zig-zag, Boustrophedon
The Walls	White	Side Blocks	6 chars	Vertical/Horizontal blocks
The Center	Purple	Inner Core	4 chars	Radiating from Center

Table 3: Selected Vocabulary Frequency (Thematic)

Character	Pinyin	Meaning	Frequency/Context
心	Xin	Heart/Mind	Center axis; emotional core
思	Si	Longing/Think	Ubiquitous; the engine of the poem
傷	Shang	Wound/Sad	Emotional damage
歸	Gui	Return	The central plea to the husband
離	Li	Separation	The state of current reality
神	Shen	Spirit/God	Used in "Heaven/Spirit" judgements
蘇	Su	Su (Surname)	Embedded in the

Character	Pinyin	Meaning	Frequency/Context
			central axis (Signature)

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